

Diabetes Analysis - Tuning Experiments

Overview

This notebook contains preprocessing, feature engineering, and hyperparameter tuning experiments on the UC Irvine *Diabetes 130-US Hospitals for Years 1999-2008* dataset. The purpose of this analysis is to determine the pipeline hyperparameters that will maximize the recall of hospital readmission predictions, in order to capture more patients at risk of needing care within 30 days of discharge. Specifically, we will run three experiments to select among five classifier models, three sampling methods, and three feature selection techniques. The sampling experiment is especially important, as this dataset is heavily imbalanced.

```
In [7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
```

```
In [9]: import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
warnings.simplefilter(action='ignore', category=UserWarning)
```

Load Dataset

```
In [39]: path = "../data/diabetic_data.csv"
df = pd.read_csv(path, low_memory=False)

with pd.option_context('display.max_columns', None):
    display(df.head())
```

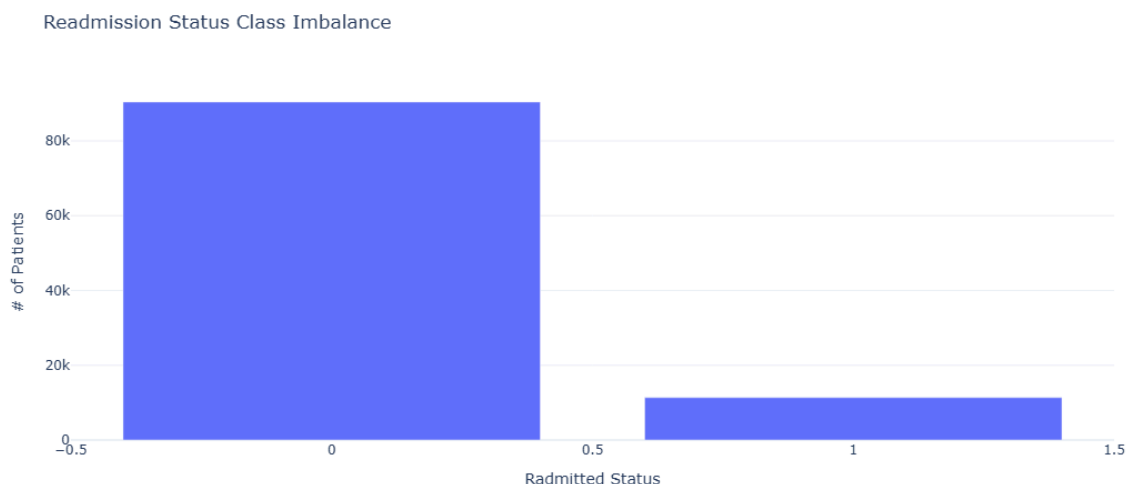
	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	disch
0	2278392	8222157	Caucasian	Female	[0-10)	?		6
1	149190	55629189	Caucasian	Female	[10-20)	?		1
2	64410	86047875	AfricanAmerican	Female	[20-30)	?		1
3	500364	82442376	Caucasian	Male	[30-40)	?		1
4	16680	42519267	Caucasian	Male	[40-50)	?		1

Data Cleaning

This section repeats the steps taken in our preprocessing notebook. The first of these is to convert the target column, *readmitted*, to a binary (1 if readmitted within 30 days, 0 otherwise).

```
In [42]: df["readmitted"] = df["readmitted"].map({
    "<30": 1, # readmitted before 30 days
    ">30": 0, # readmitted after 30 days
    "NO": 0
})

#Plots the count of each target class value
fig = px.bar(df['readmitted'].value_counts(), template = 'plotly_white', height = 500,
             title = 'Readmission Status Class Imbalance')
fig.update_layout(showlegend = False, xaxis_title = 'Radmitted Status', yaxis_title = '# of Patients')
fig.show()
```



The positive class represents only 10% of the overall data. We will need to be careful about how we evaluate model performance: predicting 0 for all patients will result in high accuracy, but poor recall and precision.

Missing values in string features are represented as '?'. To ensure that Pandas functions for handling NA values work on these rows, we will replace the missing values with np.nan. This will allow us to properly detect the proportion of missing values across all columns.

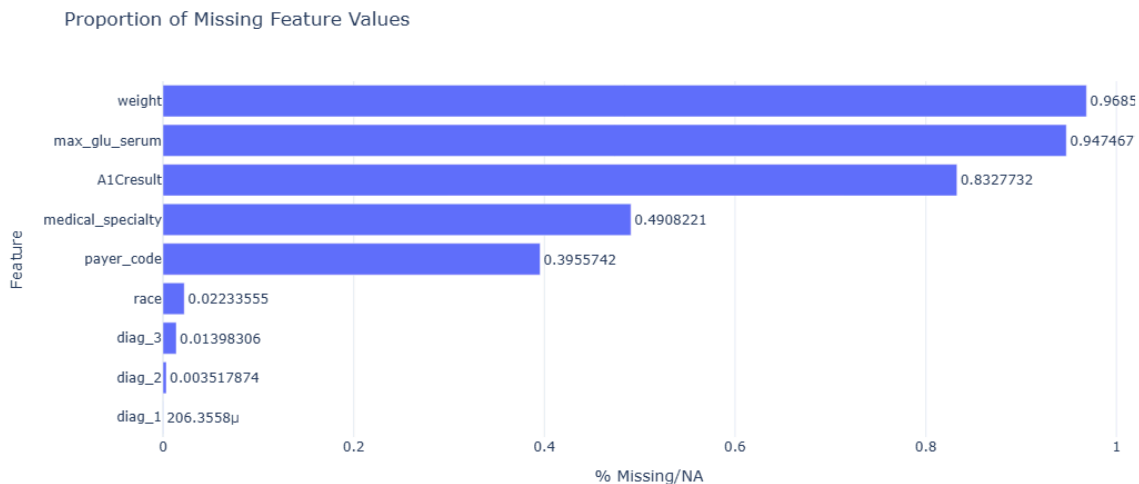
```
In [45]: #Replaces missing strings with NAN, ensuring all missing values are handled correct
df.replace('?', np.nan, inplace=True)

#Counts the proportion of NA values across all columns
na_prop = (df.isna().sum() / len(df)).to_frame(name='pct_rows')
#Displays the proportion of missing information across all features with NA values
display(na_prop[na_prop['pct_rows'] > 0].sort_values(by = 'pct_rows', ascending = F
```

	pct_rows
weight	0.968585
max_glu_serum	0.947468
A1Cresult	0.832773
medical_specialty	0.490822
payer_code	0.395574
race	0.022336
diag_3	0.013983
diag_2	0.003518
diag_1	0.000206

weight, max_glu_serum, and A1Cresult are missing too many values to be useful. We don't have any other features to calculate patient weight, such as height and BMI, so this will have to be dropped. max_glu_serum and A1Cresult will be used for feature engineering, then dropped. The rest can be imputed.

```
In [48]: fig = px.bar(na_prop[na_prop['pct_rows'] > 0], x = 'pct_rows', height = 500, width
          title = 'Proportion of Missing Feature Values', text_auto = True)
fig.update_layout(xaxis_title = '% Missing/NA', yaxis_title = 'Feature')
fig.update_traces(textposition='outside')
fig.update_yaxes(categoryorder='total ascending')
fig.show()
```



Feature Engineering

While we lack test results, we may be able to glean useful information from knowing that a patient received a blood test. This may indicate the level of care they received, or the state of their condition that required further testing. For that reason, we will convert these features into binary features - `glu_tested` and `A1C_tested` - that measure whether or not these values are NA.

Our EDA found that `time_in_hospital` and `num_medications` are correlated with one another and positively correlated with the target feature. We will combine these into an engineered feature, `severity`, that measures the state of a patient's condition during their hospitalization.

```
In [52]: df['glu_tested'] = ~df['max_glu_serum'].isna()
df['A1C_tested'] = ~df['A1Cresult'].isna()

df['severity'] = df['time_in_hospital'] * (df['num_medications'] + 1)
```

There are over 700 unique ICD9 codes across the diagnosis columns (`diag_1`, `diag_2`, `diag_3`). We will be using one-hot encoding to preprocess these features, and doing so with so many unique values will result in hundreds of sparse features. Therefore, we will group each ICD9 code into one of nine conditions, reducing the size of our feature space.

```
In [55]: def match_icd9_group(diag_col):
    #ICD9 codes referring to an external cause begin with a letter E-V. I'll replace
    diag_col = diag_col.str.replace(r'[E-V]\d*', '1000', regex = True).astype(float)

    #Matches each condition with its ICD9 grouping
    conditions = [
        (diag_col.between(390, 459)) | (diag_col == 785), #circulatory
        (diag_col.between(460, 519)) | (diag_col == 786), #respiratory
        (diag_col.between(520, 579)) | (diag_col == 787), #digestive
        np.floor(diag_col) == 250, #diabetes
        diag_col.between(800, 999), #injury
```

```

diag_col.between(710, 739), #musculoskeletal
(diag_col.between(580, 629)) | (diag_col == 788), #genitourinary
diag_col.between(140, 239), #neoplasms (cancer/other tissue growths)
diag_col.isna() #na
] #anything else is defined as 'other'

groups = [
    'circulatory',
    'respiratory',
    'digestive',
    'diabetes',
    'injury',
    'musculoskeletal',
    'genitourinary',
    'neoplasms',
    'na'
]

return pd.Series(np.select(conditions, groups, default = 'other'))

#Maps each diagnosis code to its ICD9 group
df['diag_1g'] = match_icd9_group(df['diag_1'])
df['diag_2g'] = match_icd9_group(df['diag_2'])
df['diag_3g'] = match_icd9_group(df['diag_3'])
df['diag_1g'].value_counts(normalize = True)

```

```

Out[55]: diag_1g
circulatory      0.299088
other            0.178567
respiratory      0.141727
digestive        0.093106
diabetes         0.086050
injury           0.068530
genitourinary    0.050282
musculoskeletal  0.048710
neoplasms        0.033734
na               0.000206
Name: proportion, dtype: float64

```

The medical_specialty feature poses a similar problem. Prior to one-hot encoding, we will group the 72 unique values to one of 8 'groups'.

```

In [59]: specialty_map = {
    # Internal Medicine
    'InternalMedicine': 'Internal_Medicine',
    'Nephrology': 'Internal_Medicine',
    'Pulmonology': 'Internal_Medicine',
    'Gastroenterology': 'Internal_Medicine',
    'Cardiology': 'Internal_Medicine',

    # General
    'Family/GeneralPractice': 'General',

    # Emergency
    'Emergency/Trauma': 'Emergency',

```

```

# Surgery
'Surgery-Cardiovascular/Thoracic': 'Surgery',
'Surgery-General': 'Surgery',
'Orthopedics': 'Surgery',
'Surgery-Vascular': 'Surgery',
'Urology': 'Surgery',
'Orthopedics-Reconstructive': 'Surgery',

# Pysch
'Psychiatry': 'Pysch',

# Reproductive Health
'ObstetricsandGynecology': 'OB-GYN',

# Diagnosis
'Radiologist': 'Diagnosis',

#other
'Other': 'Other',
}

df['specialty_grouped'] = df['medical_specialty'].map(specialty_map).fillna('Unknown')
df.drop('medical_specialty', axis=1, inplace=True)

```

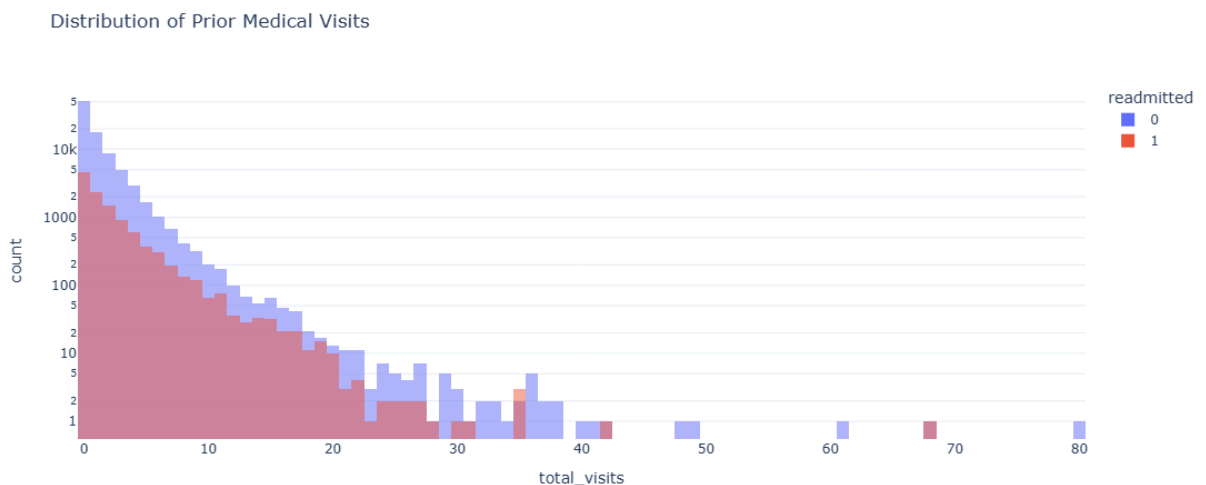
Multiple columns refer to the number of medical appointments and hospital visits a patient has had prior to their most recent hospitalization. We will aggregate these to understand the relationship between visits and risk of readmission.

```

In [65]: df['total_visits'] = df['number_inpatient'] + df['number_outpatient'] + df['number_

fig = px.histogram(df, color = 'readmitted', x = 'total_visits', barmode = 'overlay',
                  height = 500, width = 800, title = 'Distribution of Prior Medical
fig.show()

```



Splitting Data

```
In [68]: from sklearn.model_selection import train_test_split

np.random.seed(631)

#Columns are dropped to avoid data leakage, to reduce the size of our feature space
X = df.drop(columns = ["encounter_id", "patient_nbr", "weight", "max_glu_serum", "A
y = df["readmitted"]

#We stratify on the target due to the class imbalance of our dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y
```

Pipeline Initialization

```
In [166... from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer, make_column_selector
from sklearn.preprocessing import OneHotEncoder
from sklearn.feature_selection import VarianceThreshold
from imblearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier

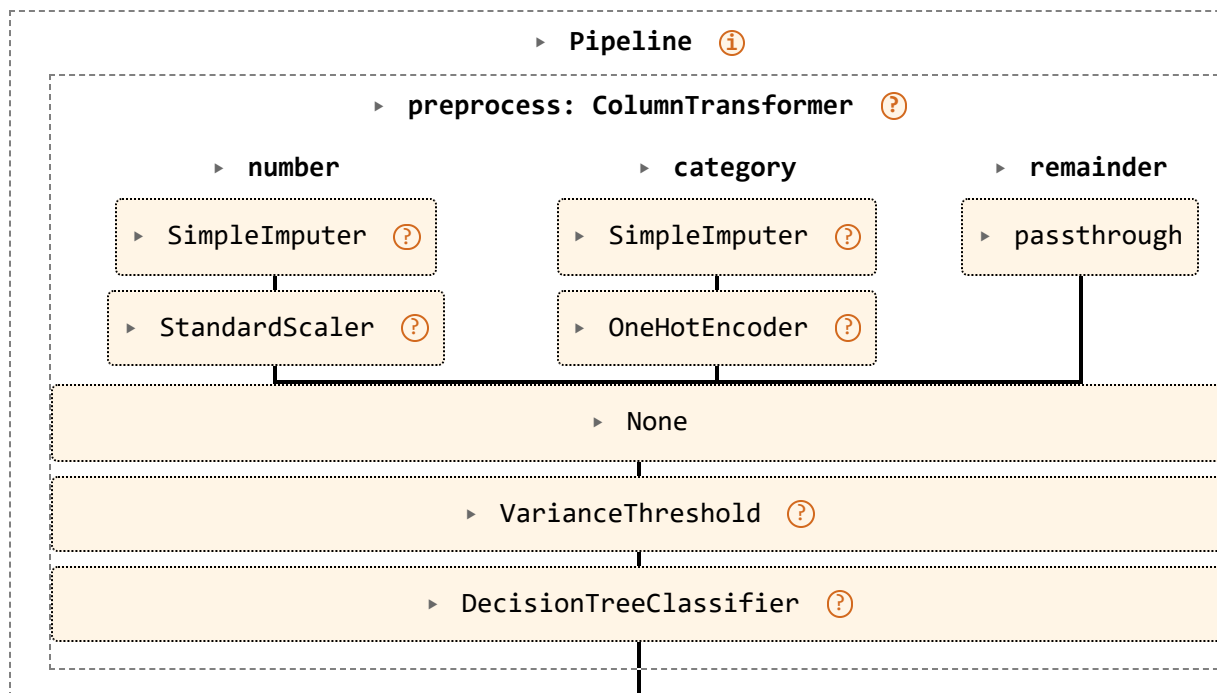
#Missing numeric values are replaced with column medians, then the values are scale
numeric_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy = 'median')),
    ('scaler', StandardScaler())
])

#Missing categorical columns are replaced with the most frequent value, then conver
categorical_transformer = Pipeline([
    ('imputer', SimpleImputer(strategy = 'most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown = 'ignore'))
])

#Some features are booleans. These are already in the range of [0,1], so they do no
prep = ColumnTransformer([
    ('number', numeric_transformer, make_column_selector(dtype_include='number')),
    ('category', categorical_transformer, make_column_selector(dtype_include='objec
], remainder = 'passthrough')

#The pipeline preprocesses the data, samples to address class imbalance, selects th
clf_pipeline = Pipeline([
    ('preprocess', prep),
    ('sampler', None),
    ('selector', VarianceThreshold(threshold = 0.3)),
    ('clf', DecisionTreeClassifier())
])

display(clf_pipeline)
```



Experiment 1: Model Selection

In this experiment, we will test five different classifiers on this dataset:

- Logistic Regression
- Decision Tree
- K-Nearest Neighbors
- Random Forest
- AdaBoost

We expect that the ensemble methods, Random Forest and AdaBoost, will perform best. Random Forest can learn nonlinear features better than linear models like Logistic Regression and generalize better than a single Decision Tree. The boosting technique of AdaBoost makes it better at classifying less frequent samples, reducing the impact of class imbalance on model performance. All model configurations will be tested using undersampled data, with features selected using a variance threshold of 0.3.

Tuning

```

In [170... from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier

#Defines the metrics each model will be evaluated on
scoring = ['accuracy', 'precision', 'recall', 'f1', 'roc_auc']

#Defines the hyperparameter combinations we will test for each model

```

```

param_grid = [
    {
        'clf': [LogisticRegression(max_iter = 1000, solver = 'saga')],
        'clf__l1_ratio': [0, 0.5, 1],
        'clf__C': [0.01, 0.1, 1, 10],
        'clf__class_weight': [None, 'balanced']
    },
    {
        'clf': [DecisionTreeClassifier()],
        'clf__splitter': ['best', 'random'],
        'clf__max_depth': [None, 10],
        'clf__class_weight': [None, 'balanced']
    },
    {
        'clf': [KNeighborsClassifier()],
        'clf__n_neighbors': [3, 5, 7],
        'clf__weights': ['uniform', 'distance']
    },
    {
        'clf': [RandomForestClassifier()],
        'clf__n_estimators': [50, 100],
        'clf__max_depth': [None, 10],
        'clf__max_features': ['sqrt', 'log2', None],
        'clf__class_weight': [None, 'balanced']
    },
    {
        'clf': [AdaBoostClassifier()],
        'clf__n_estimators': [50, 100],
        'clf__learning_rate': [0.5, 1.0]
    }
]

```

```

#Evaluates each model configuration using 5-fold cross validation, refitting to max
grid = GridSearchCV(clf_pipeline, param_grid, cv = 5, scoring = scoring, refit = 'f
grid.fit(X_train, y_train)

```

```

#Converts the CV-scores to a pandas dataframe
scores = pd.DataFrame(grid.cv_results_)
display(scores.head())

```

```

#Persists the scores to local memory
scores.to_csv('../data/tuning/cv_scores.csv', index = False)

```

	mean_fit_time	std_fit_time	mean_score_time	std_score_time	param_
0	6.065732	2.289473	0.295458	0.003652	LogisticRegression(max_iter=10 solver='saq
1	7.106424	2.845504	0.297221	0.004123	LogisticRegression(max_iter=10 solver='saq
2	6.811462	2.777071	0.300236	0.005963	LogisticRegression(max_iter=10 solver='saq
3	1.370133	0.137268	0.306872	0.006482	LogisticRegression(max_iter=10 solver='saq
4	1.512707	0.052783	0.298055	0.005066	LogisticRegression(max_iter=10 solver='saq

5 rows × 56 columns

```
In [247... #Loads the scores from memory
scores = pd.read_csv('../data/tuning/cv_scores.csv')
#Gets the classifier name, using regex to split camel-casing into separate words ("
scores['clf'] = scores['param_clf'].str.split('(').str[0].str.replace('Classifier',

#Displays the top 5 models by average recall
with pd.option_context('display.max_colwidth', None):
    display(scores[['clf', 'params', 'mean_fit_time', 'mean_score_time', 'mean_test_
                    'mean_test_f1', 'mean_test_roc_auc']].sort_values(by = 'mean_test_r
```

	clf	params	mean_fit_time	mean_score_time	mean_test_accuracy	n
31	Decision Tree	{'clf': DecisionTreeClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__splitter': 'random'}	1.312359	0.299588	0.624969	
30	Decision Tree	{'clf': DecisionTreeClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__splitter': 'best'}	1.594046	0.297793	0.624810	
60	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 50}	24.714188	0.397552	0.685304	
61	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 100}	48.524891	0.503679	0.687638	
56	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__max_features': 'sqrt', 'clf__n_estimators': 50}	8.154035	0.398138	0.693583	

In [269...

```
#Calculates each model's average accuracy, precision, recall, and f1-score across a
best_scores = pd.melt(scores.groupby('clf').apply(lambda x: x.nlargest(1, 'mean_test_accuracy'),
id_vars = 'clf', var_name = 'metric', value_name = 'avg_score')

#Plots the average values as a bar chart, encoding model as bar color, average score as bar x
fig = px.bar(best_scores, x = 'metric', y = 'avg_score', color = 'clf', template = 'simple',
height = 600, width = 1000, text_auto = True, barmode = 'group', title = 'Average Score by Metric')
#Adds text labels above each bar
fig.update_traces(textposition='outside', texttemplate='%{y:.2f}')
fig.update_layout(yaxis_title = 'Average Score', xaxis_title = 'Metric', bargroupgap = 10)

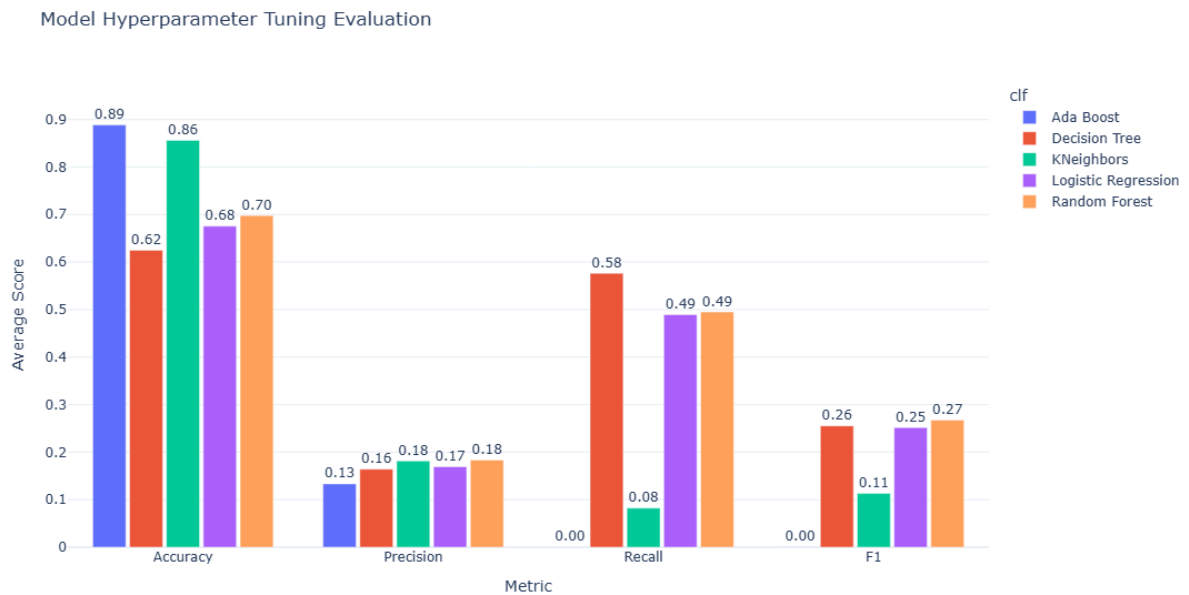
fig.update_xaxes(
tickmode = 'array',
```

```

    tickvals = ['mean_test_accuracy', 'mean_test_precision', 'mean_test_recall', 'm
    ticktext = ['Accuracy', 'Precision', 'Recall', 'F1']
)

fig.show()

```



This chart visualizes the performance metrics of the model configurations that maximize F1-score (balance precision and recall). Immediately, we can see the difficulties in training and evaluating models that result from the high class imbalance. Most patients belong to the negative class; models predict the majority class more often, resulting in misleading high-accuracy rates. A low number of readmitted patients and lack of high correlations between any features and the target explains the low precision scores. Without using any sampling techniques, we find that decision tree outperforms all other models on recall. We will investigate what impact undersampling has on these results.

```

In [218... from imblearn.under_sampling import RandomUnderSampler

#Replaces the sampling technique
clf_pipeline.set_params(sampler=RandomUnderSampler())

#Evaluates each model configuration using 5-fold cross validation, refitting to max
grid = GridSearchCV(clf_pipeline, param_grid, cv = 5, scoring = scoring, refit = 'f
grid.fit(X_train, y_train)

#Converts the CV-scores to a pandas dataframe
scores = pd.DataFrame(grid.cv_results_)
display(scores.head())

#Persists the scores to Local memory
scores.to_csv('../data/tuning/cv_scores_undersample.csv', index = False)

```

	mean_fit_time	std_fit_time	mean_score_time	std_score_time	param_
0	1.094735	0.030932	0.321161	0.010803	LogisticRegression(max_iter=10 solver='saq
1	1.160969	0.097585	0.319440	0.015806	LogisticRegression(max_iter=10 solver='saq
2	1.053793	0.029943	0.323708	0.019551	LogisticRegression(max_iter=10 solver='saq
3	1.144017	0.036477	0.326830	0.008038	LogisticRegression(max_iter=10 solver='saq
4	1.082769	0.041169	0.321152	0.007103	LogisticRegression(max_iter=10 solver='saq

5 rows × 56 columns

```
In [11]: #Loads the scores from memory
scores = pd.read_csv('../data/tuning/cv_scores_undersample.csv')
#Gets the classifier name, using regex to split camel-casing into separate words ("
scores['clf'] = scores['param_clf'].str.split('(').str[0].str.replace('Classifier',

#Displays the top 5 models by average recall
with pd.option_context('display.max_colwidth', None):
    display(scores[['clf', 'params', 'mean_fit_time', 'mean_score_time', 'mean_test_
                    'mean_test_f1', 'mean_test_roc_auc']].sort_values(by = 'mean_test_r
```

	clf	params	mean_fit_time	mean_score_time	mean_test_accuracy	n
48	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': None, 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 50}	6.310864	0.431039	0.606594	
49	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': None, 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 100}	11.640531	0.545473	0.604210	
60	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 50}	6.216246	0.430315	0.614504	
61	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': 'balanced', 'clf__max_depth': 10, 'clf__max_features': None, 'clf__n_estimators': 100}	11.406735	0.548381	0.617476	
46	Random Forest	{'clf': RandomForestClassifier(), 'clf__class_weight': None, 'clf__max_depth': 10, 'clf__max_features': 'log2', 'clf__n_estimators': 50}	2.653959	0.443562	0.614357	

It appears that on average, Random Forest models had the highest recall. These models are showing poor precision, likely due to the undersampling method used to address class imbalance. The models are trained to be more sensitive to the minority class, reducing their ability to accurately distinguish true and false positives.

```
In [14]: #Calculates each model's average accuracy, precision, recall, and f1-score across a
#avg_scores = pd.melt(scores.groupby('clf')[['mean_test_accuracy', 'mean_test_preci
#      id_vars = 'clf', var_name = 'metric', value_name = 'avg_score'])
best_scores = pd.melt(scores.groupby('clf').apply(lambda x: x.nlargest(1, 'mean_tes
      id_vars = 'clf', var_name = 'metric', value_name = 'avg_score'))
```

```

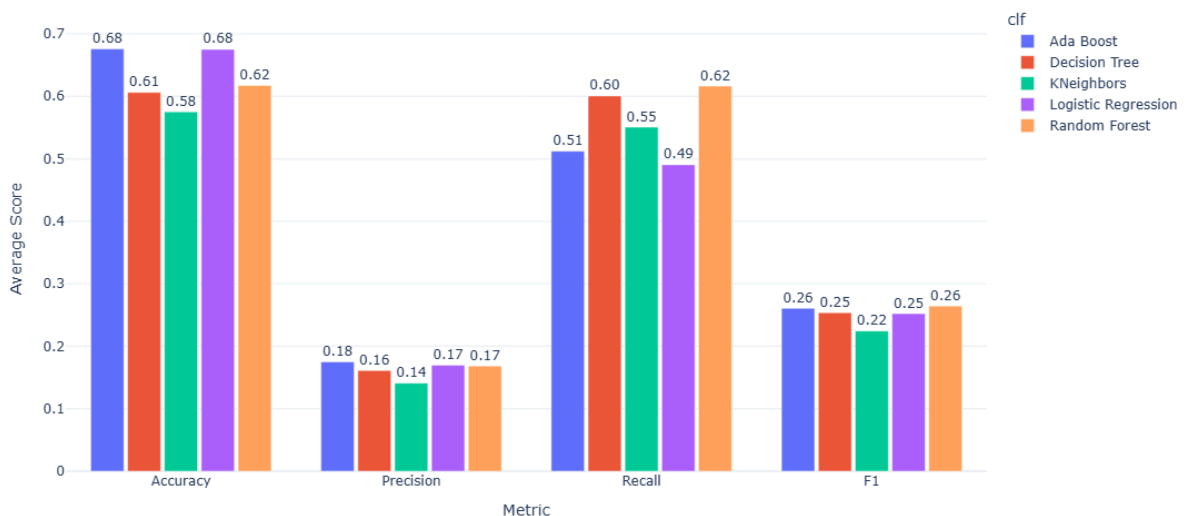
#Plots the average values as a bar chart, encoding model as bar color, average score
#and metric as bar x
fig = px.bar(best_scores, x = 'metric', y = 'avg_score', color = 'clf', template =
            height = 600, width = 1000, text_auto = True, barmode = 'group', title
#Adds text labels above each bar
fig.update_traces(textposition='outside', texttemplate='%{y:.2f}')
fig.update_layout(yaxis_title = 'Average Score', xaxis_title = 'Metric', bargroupgap=

fig.update_xaxes(
    tickmode = 'array',
    tickvals = ['mean_test_accuracy', 'mean_test_precision', 'mean_test_recall', 'm
    ticktext = ['Accuracy', 'Precision', 'Recall', 'F1']
)

fig.show()

```

Model Hyperparameter Tuning Evaluation (Undersampling)



The results are strikingly similar - high accuracy/recall and low precision/F1 scores. Notably, undersampling has improved the performance of AdaBoost and KNeighbors. All models struggle with precision, resulting in low F1-scores. While AdaBoost and Logistic Regression outperform Random Forest on accuracy, we will use Random Forest due to its high average recall.

To confirm these findings, we will train the model configurations that maximize recall, then compare the results in a precision-recall curve. This is used in place of an ROC curve, as a precision-recall curve focuses only on the positive class and is unaffected by false positives.

In [291...

```

from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier
from sklearn.metrics import precision_recall_curve

#Gets the hyperparameters for each model that maximize recall

```

```

best_recall = scores.groupby('clf').apply(lambda x: x.nlargest(1, 'mean_test_recall'))

#Splits the training data into train and validation sets
X_train_pr, X_valid_pr, y_train_pr, y_valid_pr = train_test_split(X_train, y_train,

#Initializes an empty list to store the precision-recall results
pr_res = []

for index, model, params in best_recall.reset_index()[['clf', 'params']].itertuples():
    #Initializes a dict to store the current model's results
    res = {}
    #Sets the model name attribute
    res['model'] = model

    #Updates the pipeline with the current parameters
    clf_pipeline.set_params(**eval(params))

    #Trains the pipeline and predicts using the validation set
    clf_pipeline.fit(X_train_pr, y_train_pr)
    y_prob = clf_pipeline.predict_proba(X_valid_pr)[:, 1]

    #Calculates precision and recall using the sklearn precision_recall_curve function
    precision, recall, thresholds = precision_recall_curve(y_valid_pr, y_prob)

    #Stores the results
    res['precision'] = precision
    res['recall'] = recall
    res['thresholds'] = thresholds
    pr_res.append(res)

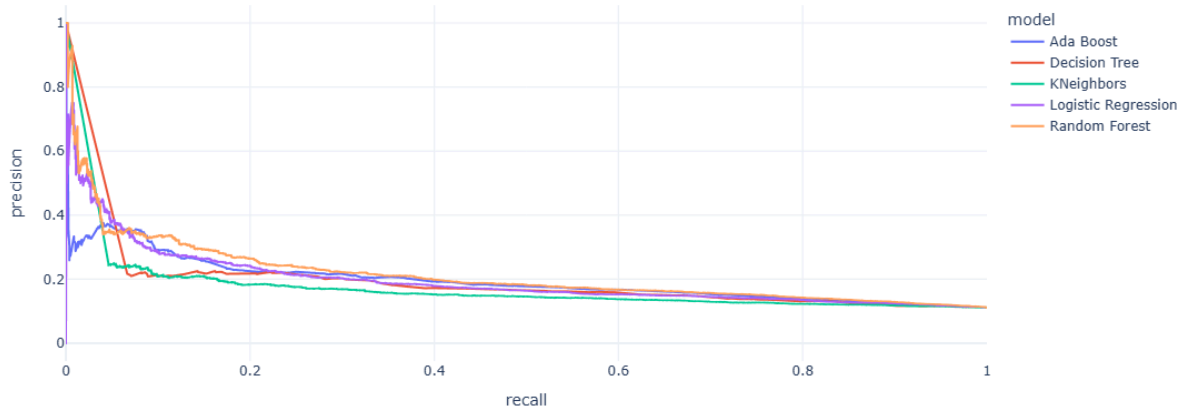
#Expands each classifier's precision and recall values from lists to individual rows
pr_frame = pd.DataFrame(pr_res).explode(['precision', 'recall'])

#Plots each model's PR curve on the same figure
fig = px.line(pr_frame, x = 'recall', y = 'precision', color = 'model', template =
              height = 500, width = 800, title = 'Model Comparison Precision-Recall')

fig.show()

```

Model Comparison Precision-Recall Curve



While the concave-upward decreasing curves on this figure emphasizes the poor precision of our classifiers, it confirms our choice of selected model. Random Forest maximizes recall with little loss in precision compared to AdaBoost and Logistic Regression.

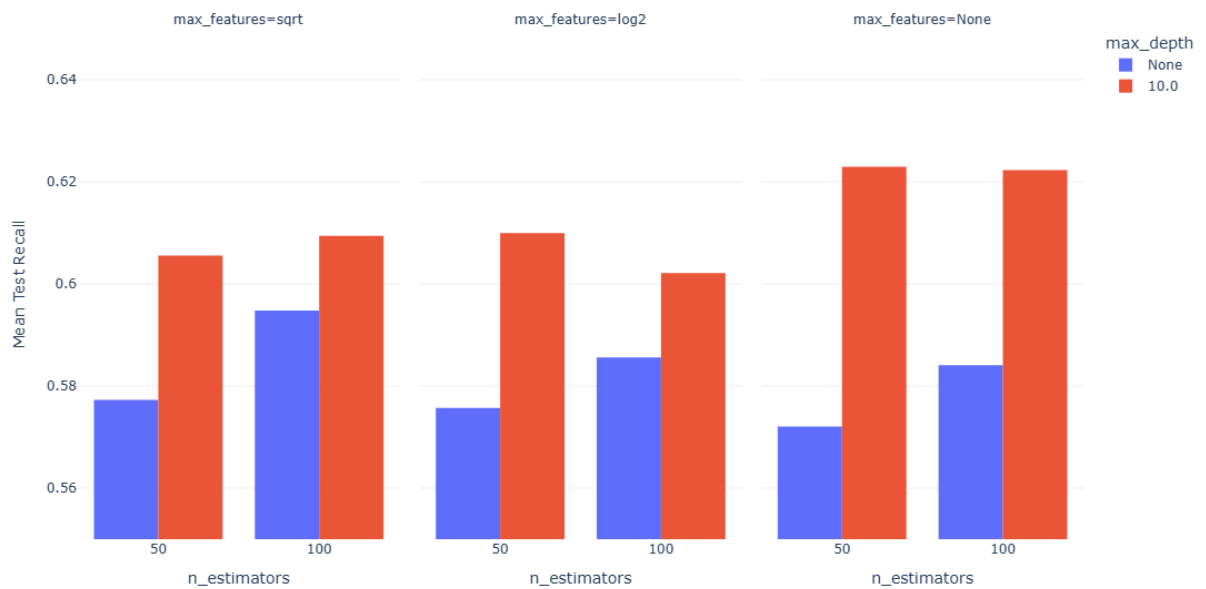
To choose the hyperparameters of our Random Forest model, we will create a visualization to compare the mean test recall across all configurations.

In [305...

```
#Retrieves the hyperparameters and recall scores of all Random Forest configuration
forest_performance = scores.loc[scores['clf'] == 'Random Forest',
    ['param_clf__n_estimators', 'param_clf__max_depth', 'param_clf__max_features'],

#Renames the hyperparameter columns to remove 'param_clf__'
forest_performance.columns = ['n_estimators', 'max_depth', 'max_features', 'mean_te
#Some hyperparameters have 'None' as an option. We'll convert them to strings to en
forest_performance['max_depth'] = forest_performance['max_depth'].fillna('None').as
forest_performance['max_features'] = forest_performance['max_features'].fillna('Non

#Plots the performance as a histogram
fig = px.histogram(forest_performance, x = 'n_estimators', y = 'mean_test_recall',
    barmode = 'group', height = 600, width = 1000, template = 'plotly
fig.update_xaxes(type = 'category')
#Limits the y-axis range to emphasize the models that maximize performance
fig.update_yaxes(range=[0.55, 0.65])
fig.update_layout(yaxis=dict(title="Mean Test Recall"))
fig.show()
```



Though recall is very similar across all model configurations, the model with `max_depth=10`, `max_features=None`, `n_estimators=50` resulted in the highest mean test recall (0.623).

Experiment 2: Sampling Method Tuning

In addition to random under sampling, we will evaluate how models perform using random over sampling and SMOTE.

In [446...

```
from imblearn.over_sampling import RandomOverSampler, SMOTE

scoring = ['accuracy', 'precision', 'recall', 'f1', 'roc_auc']

param_grid = [{
    'clf': [RandomForestClassifier(max_depth = 10, max_features = None, n_estimators = 100),
            'sampler': [RandomUnderSampler(), RandomOverSampler(), SMOTE()]
}]

#Evaluates each sampling method's performance using 5-fold cross validation, refitting on f1 over recall to improve precision without sacrificing recall
grid = GridSearchCV(clf_pipeline, param_grid, cv = 5, scoring = scoring, refit = 'f1')
grid.fit(X_train, y_train)

sampler_scores = pd.DataFrame(grid.cv_results_)
sampler_scores.to_csv('../data/tuning/sampler_scores.csv', index = False)
display(sampler_scores[['param_sampler', 'mean_fit_time', 'mean_score_time', 'mean_
```

	param_sampler	mean_fit_time	mean_score_time	mean_test_accuracy	mean_test_pre
0	RandomUnderSampler()	7.167765	0.443074	0.616863	0.616863
1	RandomOverSampler()	59.754114	0.441349	0.654031	0.654031
2	SMOTE()	59.345580	0.446587	0.869061	0.869061

```
In [40]: sampler_scores = pd.read_csv('../data/tuning/sampler_scores.csv')
sampler_scores['param_sampler'] = sampler_scores['param_sampler'].str.replace('()', '')

#Calculates each sampling method's average accuracy, precision, recall, and f1-score
avg_scores = pd.melt(sampler_scores.groupby('param_sampler').apply(lambda x: x.nlargest(
    id_vars = 'param_sampler', var_name = 'metric', value_name = 'avg_score'))

#Plots the average scores as a bar chart
fig = px.bar(avg_scores, x = 'metric', y = 'avg_score', color = 'param_sampler', template=
    height = 600, width = 1000, barmode = 'group', title = 'Sampling Technique Performance Evaluation')

fig.update_layout(xaxis_title = 'Metric', yaxis_title = 'Average Score', bargroupgap = 10)

fig.update_xaxes(
    tickmode = 'array',
    tickvals = ['mean_test_accuracy', 'mean_test_precision', 'mean_test_recall', 'mean_test_f1'],
    ticktext = ['Accuracy', 'Precision', 'Recall', 'F1']
)

fig.for_each_annotation(lambda a: a.update(text=a.text.split("mean_test_")[-1].title()))

fig.show()
```



Undersampling and oversampling resulted in similar performance, with undersampling having lower fit times and higher recall. While SMOTE greatly improved model accuracy and

precision, this technique resulted in much lower recall. Therefore, we will continue using random under sampling to address the class imbalance during training.

Experiment 3: Feature Selection Tuning

In [151...

```
from sklearn.feature_selection import SelectKBest, SelectFromModel
from sklearn.svm import LinearSVC

scoring = ['accuracy', 'precision', 'recall', 'f1', 'roc_auc']

param_grid = [
    {
        'clf': [RandomForestClassifier(max_depth = 10, max_features = None, n_estim
        'selector': [VarianceThreshold()],
        'selector__threshold': [0.1, 0.3, 0.5]
    },
    {
        'clf': [RandomForestClassifier(max_depth = 10, max_features = None, n_estim
        'selector': [SelectKBest()],
        'selector__k': ['all', 10, 20]
    },
    {
        'clf': [RandomForestClassifier(max_depth = 10, max_features = None, n_estim
        'selector': [SelectFromModel(LinearSVC(C=0.01, random_state=631, max_iter=1
    }
]

#Evaluates each feature selection technique's performance using 5-fold cross valida
#Here, we refit on f1 over recall to improve precision without sacrificing recall p
grid = GridSearchCV(clf_pipeline, param_grid, cv = 5, scoring = scoring, refit = 'f
grid.fit(X_train, y_train)

selection_scores = pd.DataFrame(grid.cv_results_)
selection_scores.to_csv('../data/tuning/selection_scores.csv', index = False)
display(selection_scores[['param_selector', 'mean_fit_time', 'mean_score_time', 'me
```

[illegible]

	param_selector	mean_fit_time	mean_score_time	mean_test_ac
1	VarianceThreshold()	5.923689	0.426395	0.6
2	VarianceThreshold()	5.961763	0.450708	0.6
5	SelectKBest()	6.116403	0.437019	0.6
6	SelectFromModel(estimator=LinearSVC(C=0.01, ma...	8.644398	0.466029	0.6
3	SelectKBest()	15.274362	0.533500	0.6

```
In [36]: selector_scores = pd.read_csv('../data/tuning/selection_scores.csv')
selector_scores['param_selector'] = selector_scores['param_selector'].str.split('(')

#Calculates each sampling method's average accuracy, precision, recall, and f1-score
avg_scores = pd.melt(selector_scores.groupby('param_selector').apply(lambda x: x.nlargest(
    id_vars = 'param_selector', var_name = 'metric', value_name = 'avg_score'))

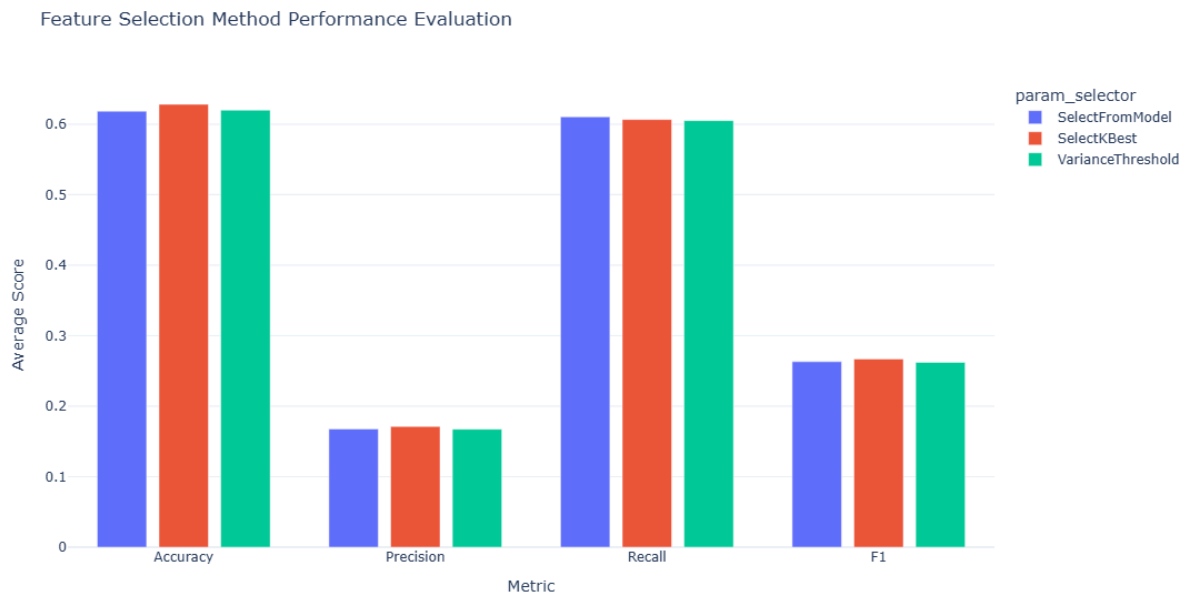
#Plots the average scores as a bar chart
fig = px.bar(avg_scores, x = 'metric', y = 'avg_score', color = 'param_selector', title = 'Feature Selection
    height = 600, width = 1000, barmode = 'group', title = 'Feature Selection

fig.update_layout(xaxis_title = 'Metric', yaxis_title = 'Average Score', bargroupgap = 10)

fig.update_xaxes(
    tickmode = 'array',
    tickvals = ['mean_test_accuracy', 'mean_test_precision', 'mean_test_recall', 'mean_test_f1'],
    ticktext = ['Accuracy', 'Precision', 'Recall', 'F1']
)

fig.for_each_annotation(lambda a: a.update(text=a.text.split("mean_test_")[-1].title))

fig.show()
```



Our findings indicate that the feature selection technique has little-to-no impact on model performance. Variance Threshold has the highest average recall and a reduced mean fit time compared to Select-From-Model, so we will use that technique in our final pipeline.

Final Evaluation

All that is left is to train the model using the tuned pipeline, then evaluate on the test set.

```
In [161... from sklearn.model_selection import cross_validate
from sklearn.metrics import accuracy_score, classification_report

tuned_pipe = Pipeline([
    ('preprocess', prep),
    ('sampler', RandomUnderSampler()),
    ('selector', VarianceThreshold(threshold = 0.3)),
    ('clf', RandomForestClassifier(max_depth = 10, n_estimators = 50, max_features
)])

#Fits the pipeline on the training set
tuned_pipe.fit(X_train, y_train)

#Evaluates on the test set
y_pred = tuned_pipe.predict(X_test)

#Prints the classification report
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.93	0.59	0.72	18083
1	0.17	0.65	0.26	2271
accuracy			0.59	20354
macro avg	0.55	0.62	0.49	20354
weighted avg	0.85	0.59	0.67	20354

The results indicate higher performance on the majority class and poor performance on predicting the minority (positive) class. Though our dataset is highly imbalanced, the importance of the minority class means that the macro average metrics provide a better understanding of overall performance. Despite our tuning experiments, the precision and recall scores are well below the expected performance for medical machine learning models.

Below, we include a feature importance plot to visualize how our model makes its predictions.

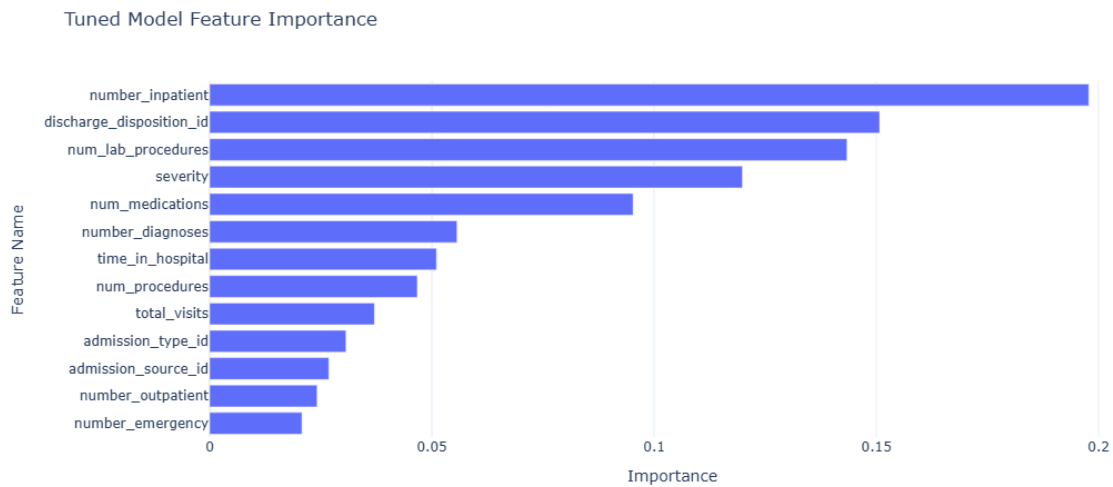
In [163...

```
#Gets the feature names and Random Forest importances
feature_importance = pd.DataFrame({
    'feature': tuned_pipe[:-1].get_feature_names_out(),
    'importance': tuned_pipe[-1].feature_importances_
})

#Each feature_name contains a prefix describing its preprocessing path. We store th
#the name
feature_importance['dtype'] = feature_importance['feature'].str.split('__').str[0]
feature_importance['dtype'] = feature_importance['dtype'].str.replace('remainder',

feature_importance['feature'] = feature_importance['feature'].str.split('__').str[1]

#Plots the importance of the top 15 features as a bar chart
fig = px.bar(feature_importance.nlargest(15, 'importance'), x = 'importance', y = '
            title = 'Tuned Model Feature Importance')
fig.update_layout(xaxis_title = 'Importance', yaxis_title = 'Feature Name')
fig.update_yaxes(categoryorder = 'total ascending')
fig.show()
```



Conclusion

Overall, our analysis is limited by the severe class imbalance of the target column. While our early approach focused on maximizing recall, later experiments that aimed to improve precision by refitting on F1-score found little improvement to either precision or recall. Further experimentation with random oversampling and SMOTE found that any improvement to precision resulted in high loss of recall.

A future analysis could expand our work by exploring additional hyperparameters or using threshold tuning techniques to balance precision and recall. We would also recommend collecting additional data on patients, as the features explored in our analysis have either low or no correlation with the target class. The inclusion of patient physical characteristics (height/weight/bmi), lab test results, and other health information may improve performance on prediction readmission.